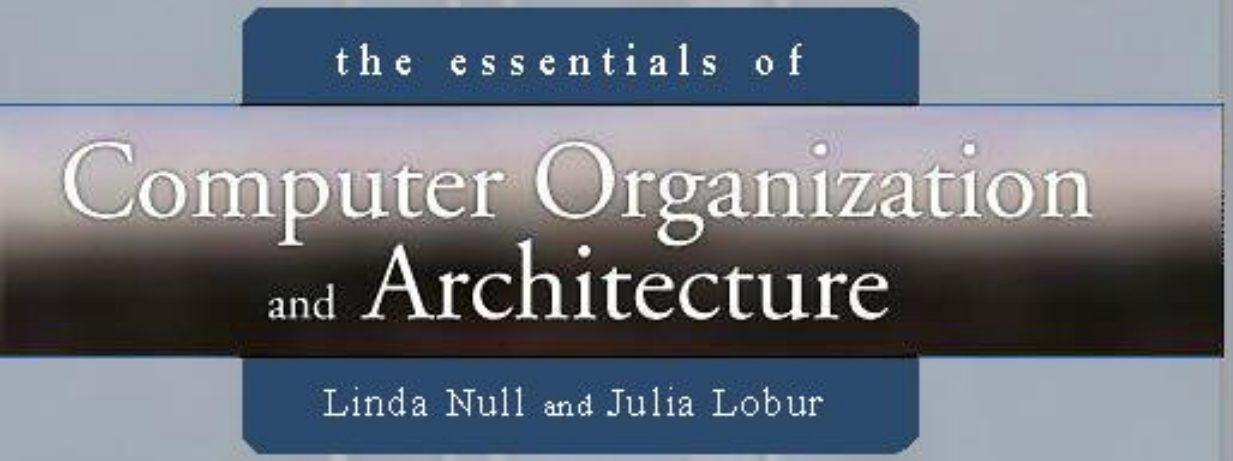


Chapter 4

MARIE: An Introduction to a Simple Computer

Chapter 4 Objectives

- Learn the components common to every modern computer system.
- Be able to explain how each component contributes to program execution.
- Understand a simple architecture invented to illuminate these basic concepts, and how it relates to some real architectures.
- Know how the program assembly process works.

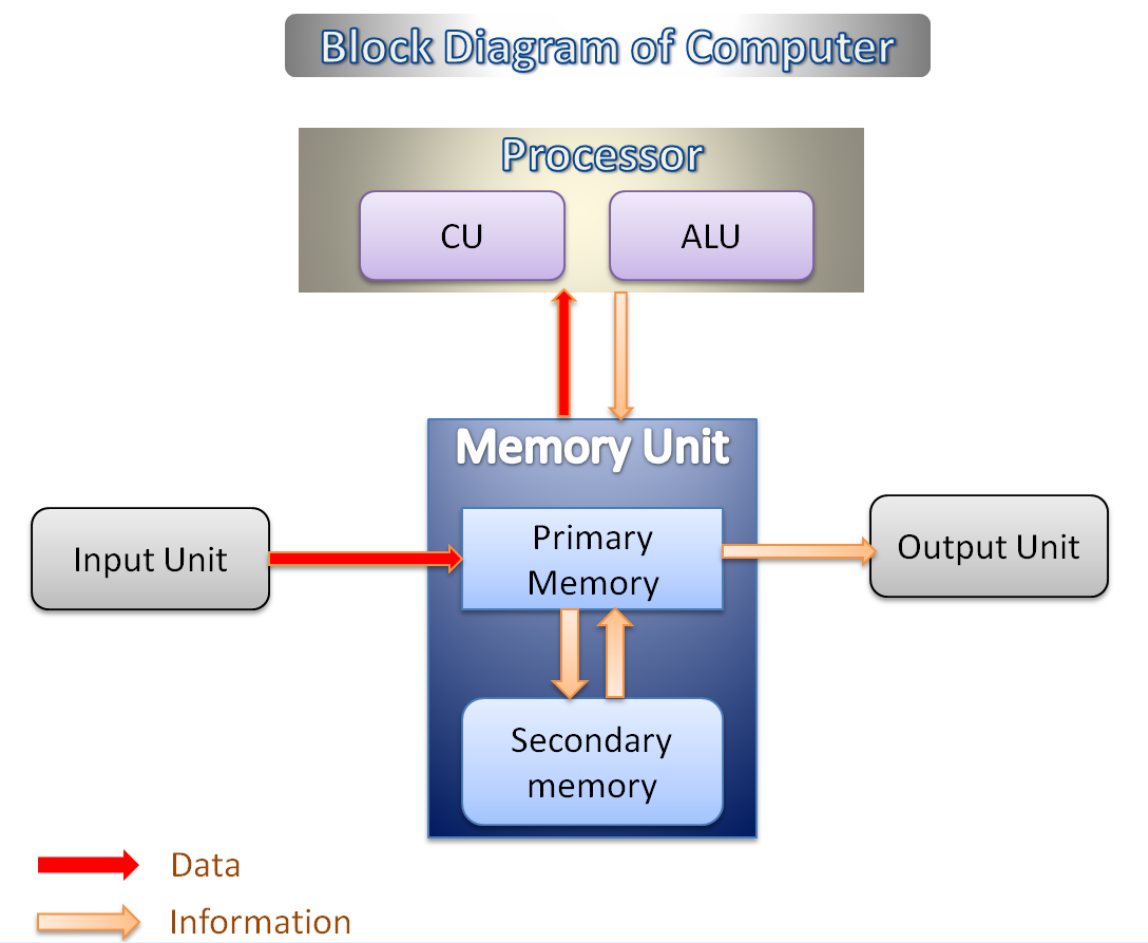
4.1 Introduction

- Chapter 1 presented a general overview of computer systems.
- In Chapter 2, we discussed how data is stored and manipulated by various computer system components.
- Chapter 3 described the fundamental components of digital circuits.
- Having this background, we can now understand how computer components work, and how they fit together to create useful computer systems.

4.1 Introduction

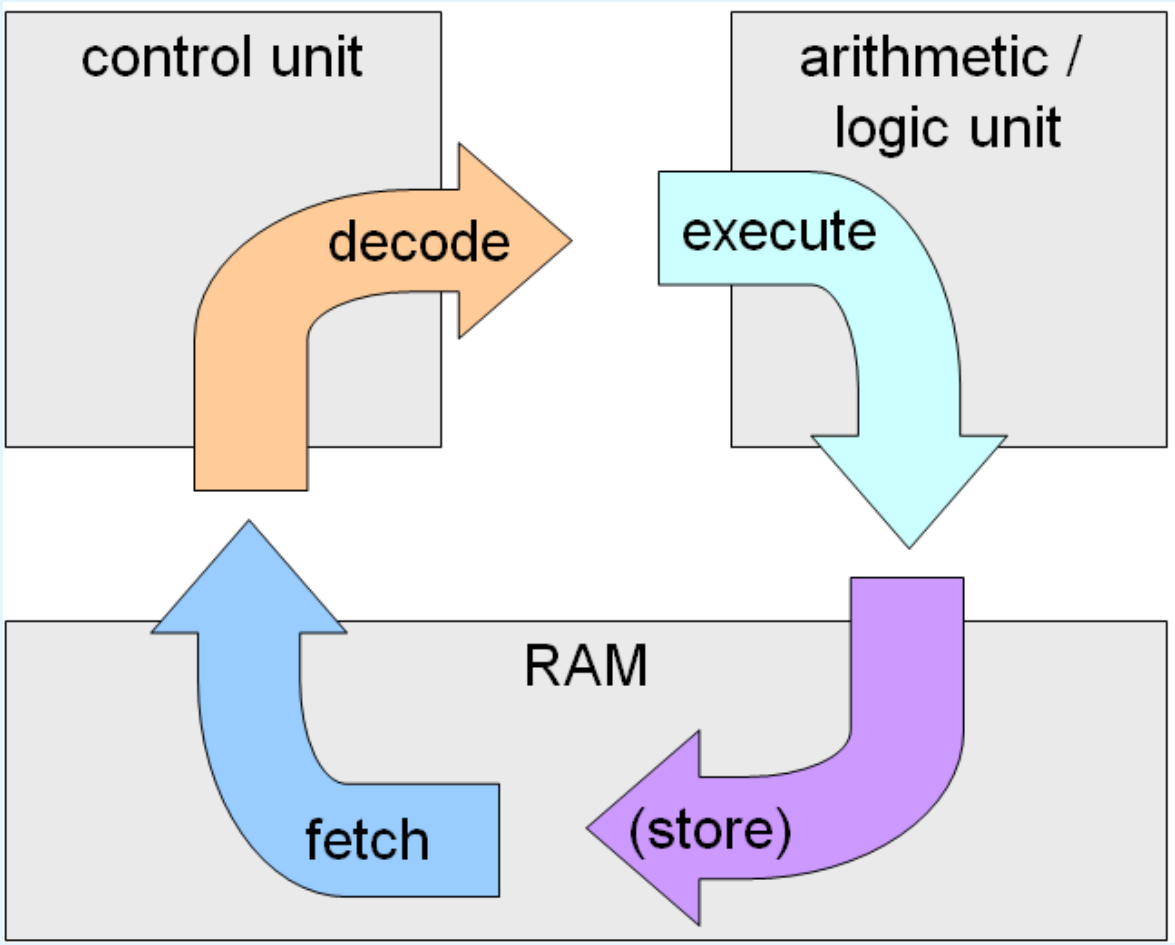
- The computer's CPU fetches, decodes, and executes program instructions.
- The two principal parts of the CPU are the *datapath* and the *control unit*.
  - The datapath consists of an arithmetic-logic unit and storage units (registers) that are interconnected by a data bus that is also connected to main memory.
  - Various CPU components perform sequenced operations according to signals provided by its control unit.

4.1 Introduction



Computer Diagram

4.1 Introduction



The machine instruction cycle

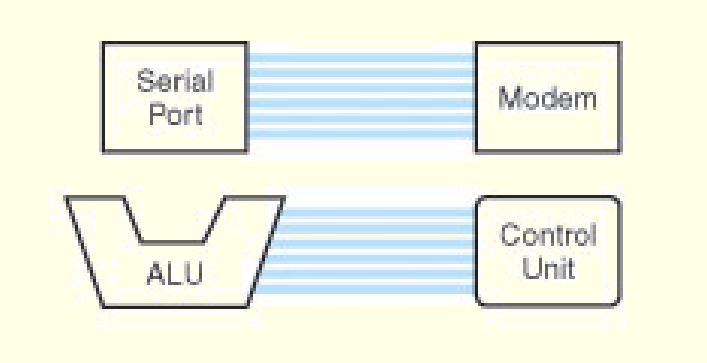
4.1 Introduction

- Registers hold data that can be readily accessed by the CPU.
- They can be implemented using D flip-flops.
  - A 32-bit register requires 32 D flip-flops.
- The arithmetic-logic unit (ALU) carries out logical and arithmetic operations as directed by the control unit.
- The control unit determines which actions to carry out according to the values in a program counter register and a status register.

4.1 Introduction

- The CPU shares data with other system components by way of a data bus.
  - A bus is a set of wires that simultaneously convey a single bit along each line.
- Two types of buses are commonly found in computer systems: *point-to-point*, and *multipoint* buses.

This is a point-to-point bus configuration:

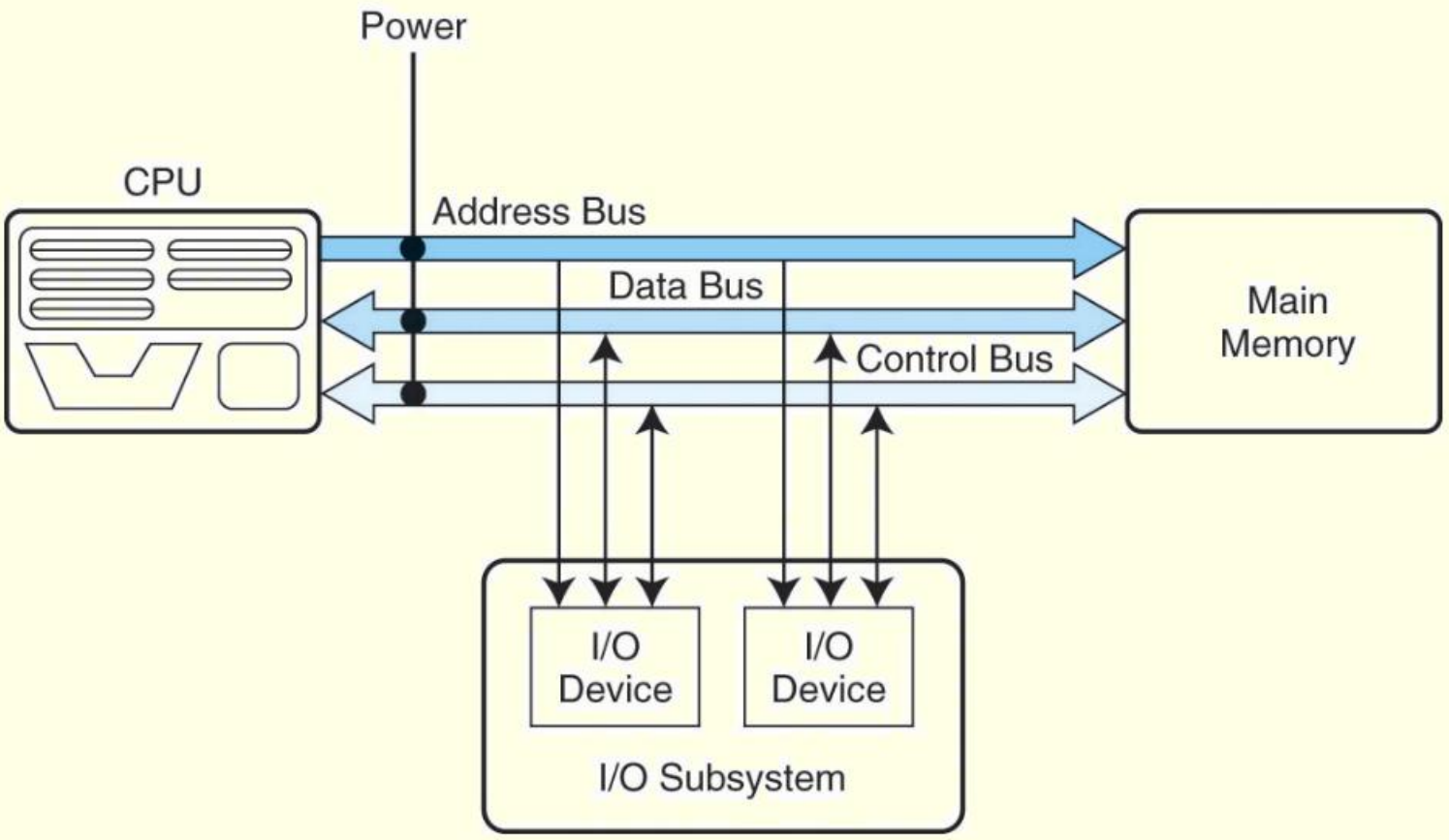


4.1 Introduction

- Buses consist of data lines, control lines, and address lines.
- While the data lines convey bits from one device to another, control lines determine the direction of data flow, and when each device can access the bus.
- Address lines determine the location of the source or destination of the data.

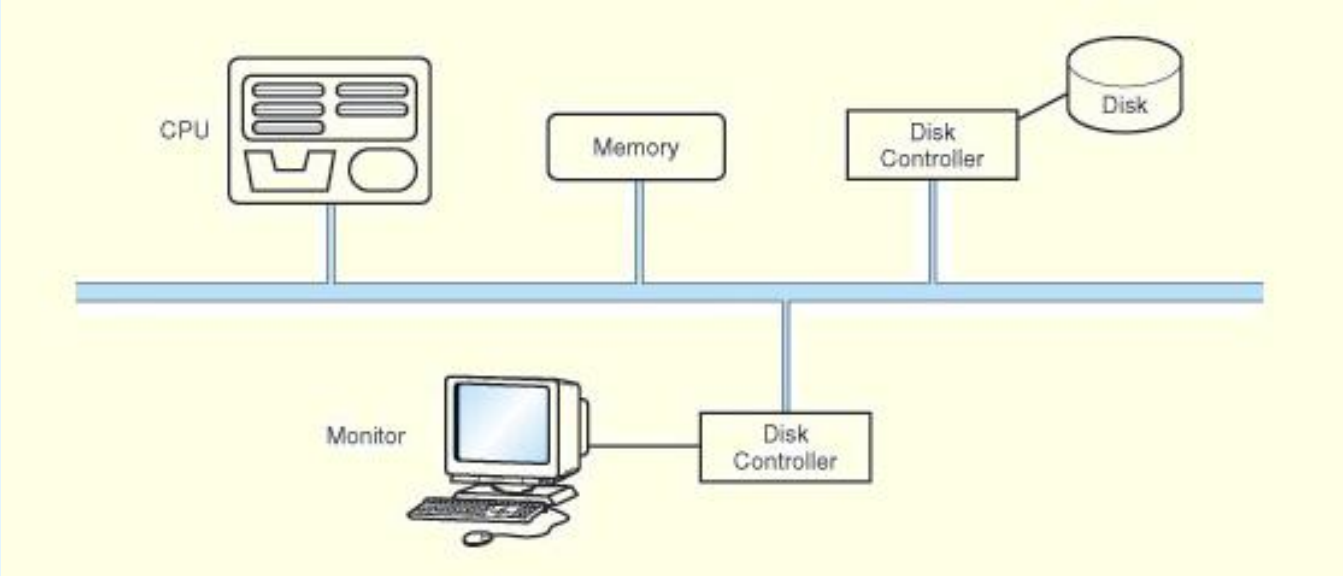
The next slide shows a model bus configuration.

4.1 Introduction



4.1 Introduction

- A multipoint bus is shown below.
- Because a multipoint bus is a shared resource, access to it is controlled through protocols, which are built into the hardware.



4.1 Introduction

- In a master-slave configuration, where more than one device can be the bus master, concurrent bus master requests must be arbitrated.
- Four categories of bus arbitration are:
  - **Daisy chain:** Permissions are passed from the highest-priority device to the lowest.
  - **Centralized parallel:** Each device is directly connected to an arbitration circuit.
  - **Distributed using self-detection:** Devices decide which gets the bus among themselves.
  - **Distributed using collision-detection:** Any device can try to use the bus. If its data collides with the data of another device, it tries again.

4.1 Introduction

- Every computer contains at least one clock that synchronizes the activities of its components.
- A fixed number of clock cycles are required to carry out each data movement or computational operation.
- The clock frequency, measured in megahertz or gigahertz, determines the speed with which all operations are carried out.
- Clock cycle time is the reciprocal of clock frequency.
  - An 800 MHz clock has a cycle time of 1.25 ns.

4.1 Introduction

- Clock speed should not be confused with CPU performance.
- The CPU time required to run a program is given by the general performance equation:

$$\text{CPU Time} = \frac{\text{seconds}}{\text{program}} = \frac{\text{instructions}}{\text{program}} \times \frac{\text{avg. cycles}}{\text{instruction}} \times \frac{\text{seconds}}{\text{cycle}}$$

  - We see that we can improve CPU throughput when we reduce the number of instructions in a program, reduce the number of cycles per instruction, or reduce the number of nanoseconds per clock cycle.

We will return to this important equation in later chapters.

4.1 Introduction

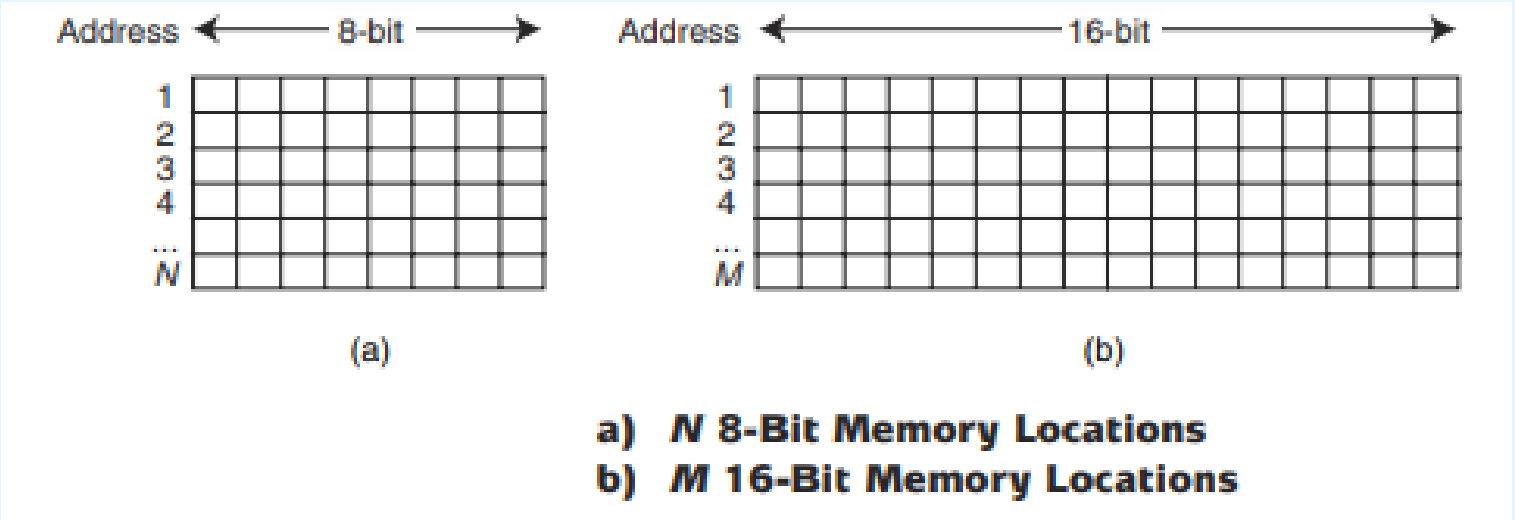
- A computer communicates with the outside world through its input/output (I/O) subsystem.
- I/O devices connect to the CPU through various interfaces.
- I/O can be memory-mapped-- where the I/O device behaves like main memory from the CPU's point of view.
- Or I/O can be instruction-based, where the CPU has a specialized I/O instruction set.

We study I/O in detail in chapter 7.

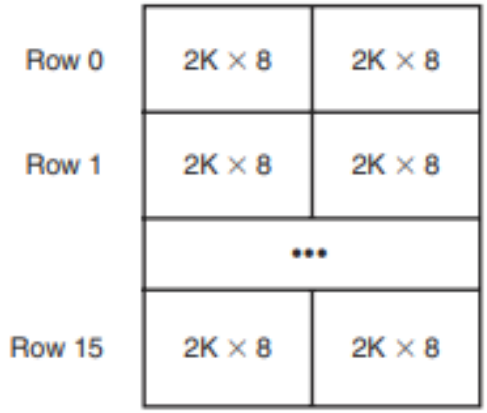
4.1 Introduction

- Computer memory consists of a linear array of addressable storage cells that are similar to registers.
- Memory can be byte-addressable, or word-addressable, where a word typically consists of two or more bytes.
- Memory is constructed of RAM chips, often referred to in terms of length × width.
- If the memory word size of the machine is 16 bits, then a 4M × 16 RAM chip gives us 4 megabytes of 16-bit memory locations.

4.1 Introduction



a) *N* 8-Bit Memory Locations
b) *M* 16-Bit Memory Locations



Memory as a Collection of RAM Chips

4.1 Introduction

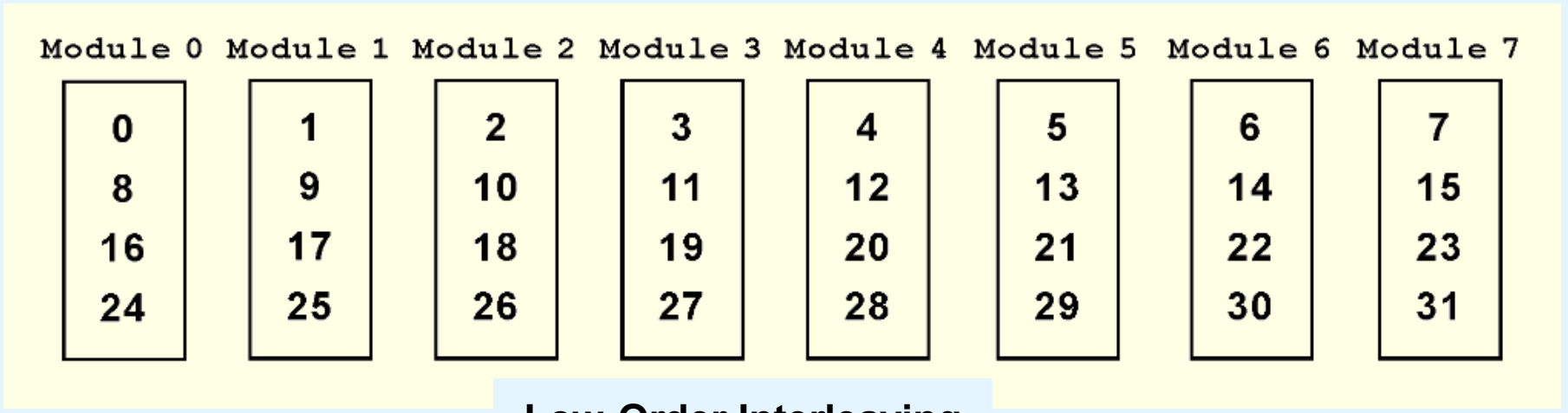
- How does the computer access a memory location corresponds to a particular address?
- We observe that 4M can be expressed as $2^2 \times 2^{20} = 2^{22}$ words.
- The memory locations for this memory are numbered 0 through 2^{22} -1.
- Thus, the memory bus of this system requires at least 22 address lines.
  - The address lines "count" from 0 to 2^{22} - 1 in binary. Each line is either "on" or "off" indicating the location of the desired memory element.

4.1 Introduction

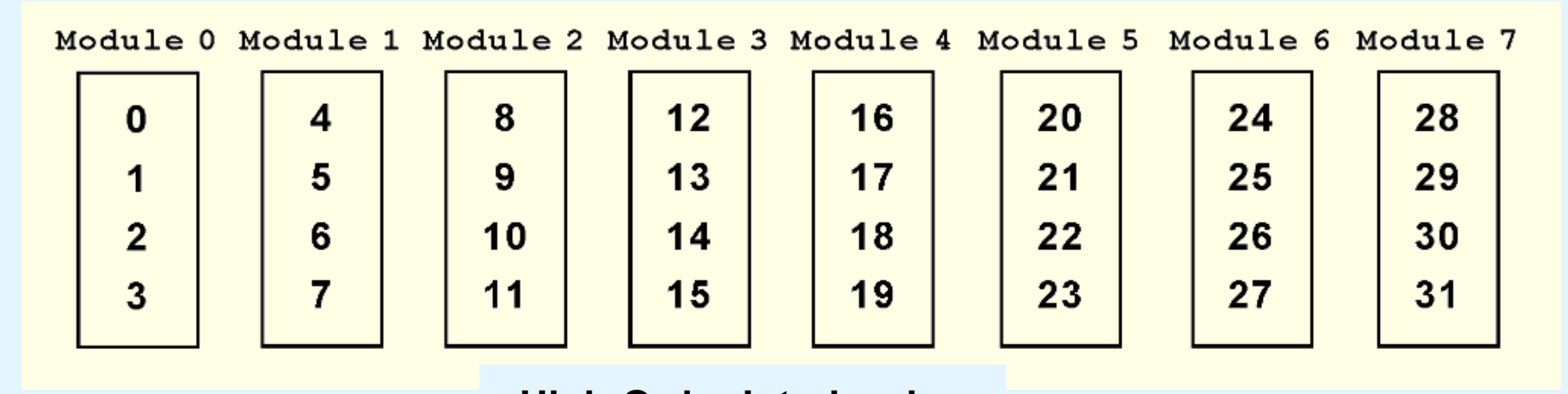
- Physical memory usually consists of more than one RAM chip.
- Access is more efficient when memory is organized into banks of chips with the addresses interleaved across the chips
- With low-order interleaving, the low order bits of the address specify which memory bank contains the address of interest.
- Accordingly, in high-order interleaving, the high order address bits specify the memory bank.

The next slide illustrates these two ideas.

4.1 Introduction



Low-Order Interleaving



High-Order Interleaving

4.1 Introduction

- The normal execution of a program is altered when an event of higher-priority occurs. The CPU is alerted to such an event through an interrupt.
- Interrupts can be triggered by I/O requests, arithmetic errors (such as division by zero), or when an invalid instruction is encountered.
- Each interrupt is associated with a procedure that directs the actions of the CPU when an interrupt occurs.
  - Nonmaskable interrupts are high-priority interrupts that cannot be ignored.

4.2 MARIE

- We can now bring together many of the ideas that we have discussed to this point using a very simple model computer.
- Our model computer, the Machine Architecture that is Really Intuitive and Easy, MARIE, was designed for the singular purpose of illustrating basic computer system concepts.
- While this system is too simple to do anything useful in the real world, a deep understanding of its functions will enable you to comprehend system architectures that are much more complex.

4.2 MARIE

The MARIE architecture has the following characteristics:

- Binary, two's complement data representation.
- Stored program, fixed word length data and instructions.
- 4K words of word-addressable main memory.
- 16-bit data words.
- 16-bit instructions, 4 for the opcode and 12 for the address.
- A 16-bit arithmetic logic unit (ALU).
- Seven registers for control and data movement.

4.2 MARIE

MARIE's seven registers are:

- Accumulator, AC, a 16-bit register that holds a conditional operator (e.g., "less than") or one operand of a two-operand instruction.
- Memory address register, MAR, a 12-bit register that holds the memory address of an instruction or the operand of an instruction.
- Memory buffer register, MBR, a 16-bit register that holds the data after its retrieval from, or before its placement in memory.

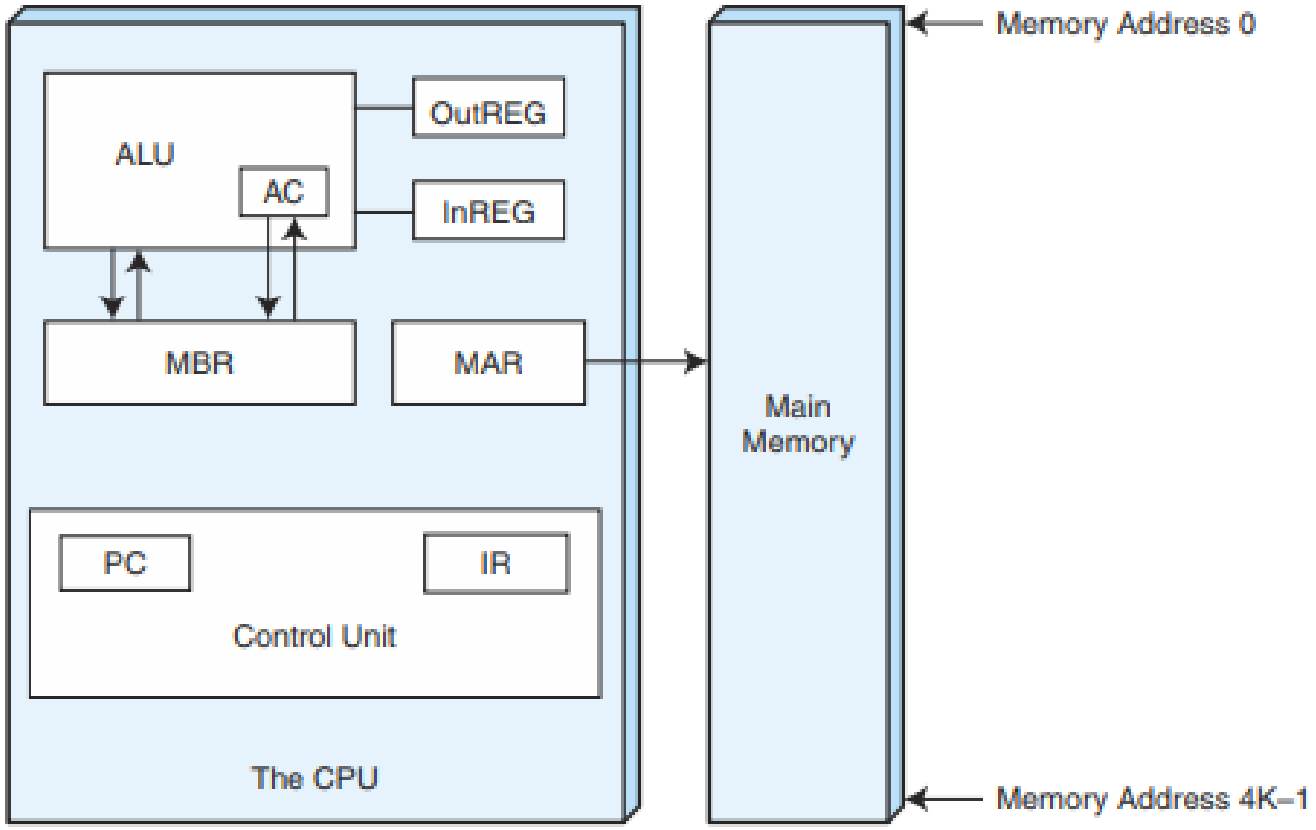
4.2 MARIE

MARIE's seven registers are:

- Program counter, PC, a 12-bit register that holds the address of the next program instruction to be executed.
- Instruction register, IR, which holds an instruction immediately preceding its execution.
- Input register, InREG, an 8-bit register that holds data read from an input device.
- Output register, OutREG, an 8-bit register, that holds data that is ready for the output device.

4.2 MARIE

This is the MARIE architecture shown graphically.



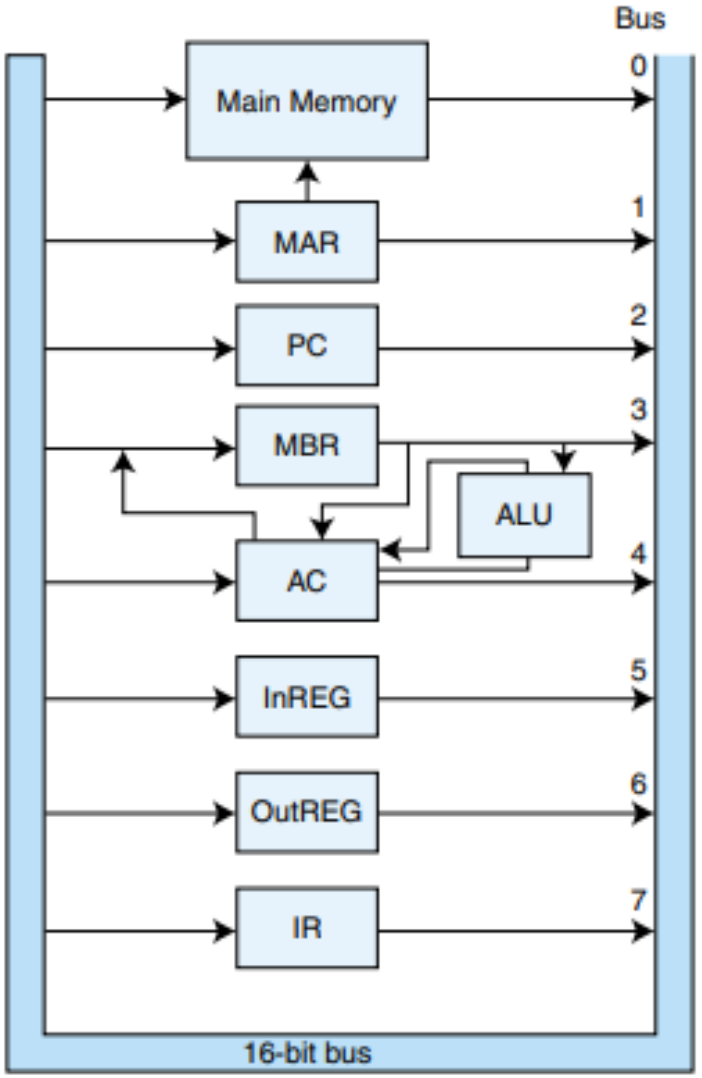
MARIE's Architecture

4.2 MARIE

- The registers are interconnected, and connected with main memory through a common data bus.
- Each device on the bus is identified by a unique number that is set on the control lines whenever that device is required to carry out an operation.
- Separate connections are also provided between the accumulator and the memory buffer register, and the ALU and the accumulator and memory buffer register.
- This permits data transfer between these devices without use of the main data bus.

4.2 MARIE

This is the MARIE data path shown graphically.



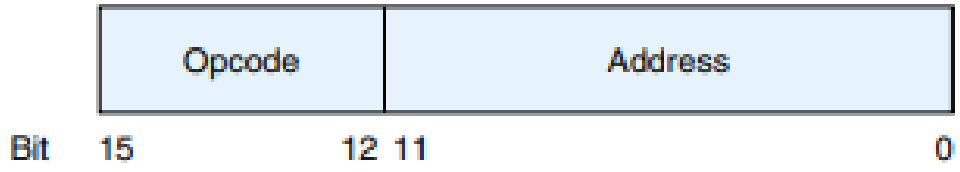
The Data Path in MARIE

4.2 MARIE

- A computer's instruction set architecture (ISA) specifies the format of its instructions and the primitive operations that the machine can perform.
- The ISA is an interface between a computer's hardware and its software.
- Some ISAs include hundreds of different instructions for processing data and controlling program execution.
- The MARIE ISA consists of only thirteen instructions.

4.2 MARIE

- This is the format of a MARIE instruction:



MARIE's Instruction Format

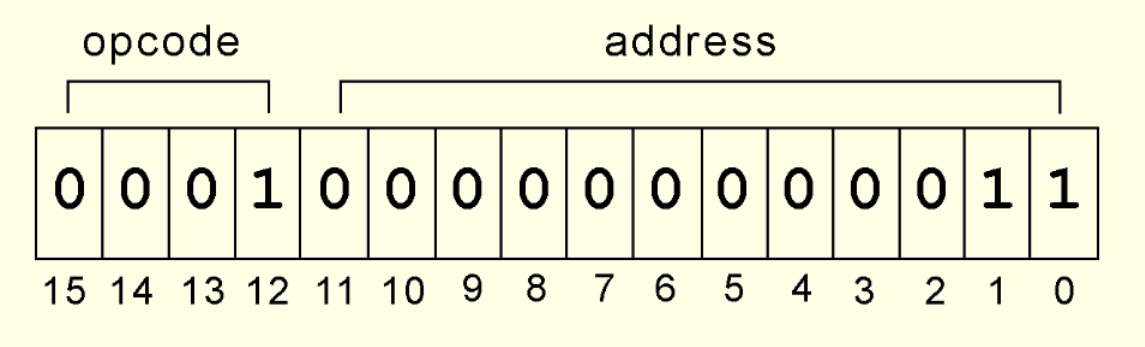
- The fundamental MARIE instructions are:

Instruction Number		Instruction	Meaning
Bin	Hex		
0001	1	Load *X*	Load the contents of address *X* into AC.
0010	2	Store *X*	Store the contents of AC at address *X*.
0011	3	Add *X*	Add the contents of address *X* to AC and store the result in AC.
0100	4	Subt *X*	Subtract the contents of address *X* from AC and store the result in AC.
0101	5	Input	Input a value from the keyboard into AC.
0110	6	Output	Output the value in AC to the display.
0111	7	Halt	Terminate the program.
1000	8	Skipcond	Skip the next instruction on condition.
1001	9	Jump *X*	Load the value of *X* into PC.

MARIE's Instruction Set

4.2 MARIE

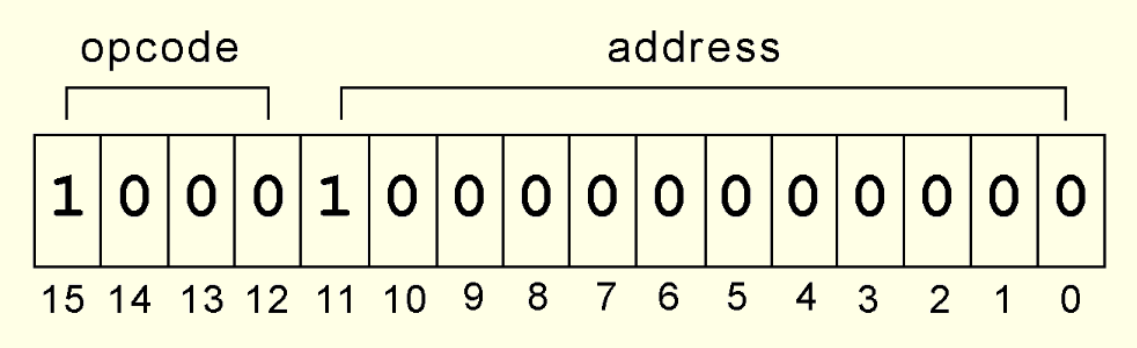
- This is a bit pattern for a **`LOAD`** instruction as it would appear in the IR:



- We see that the opcode is 1 and the address from which to load the data is 3.

4.2 MARIE

- This is a bit pattern for a **`SKIPCOND`** instruction as it would appear in the IR:



- We see that the opcode is 8 and bits 11 and 10 are 10, meaning that the next instruction will be skipped if the value in the AC is greater than zero.

What is the hexadecimal representation of this instruction?

4.2 MARIE

- Each of our instructions actually consists of a sequence of smaller instructions called *microoperations*.
- The exact sequence of microoperations that are carried out by an instruction can be specified using *register transfer language (RTL).*
- In the MARIE RTL, we use the notation M[X] to indicate the actual data value stored in memory location X, and ← to indicate the transfer of bytes to a register or memory location.

4.2 MARIE

- The RTL for the **LOAD** instruction is:

```
MAR ← X
MBR ← M[MAR], AC ← MBR
```

- Similarly, the RTL for the **ADD** instruction is:

```
MAR ← X
MBR ← M[MAR]
AC ← AC + MBR
```

4.2 MARIE

- Recall that **SKIPCOND** skips the next instruction according to the value of the AC.
- The RTL for the this instruction is the most complex in our instruction set:

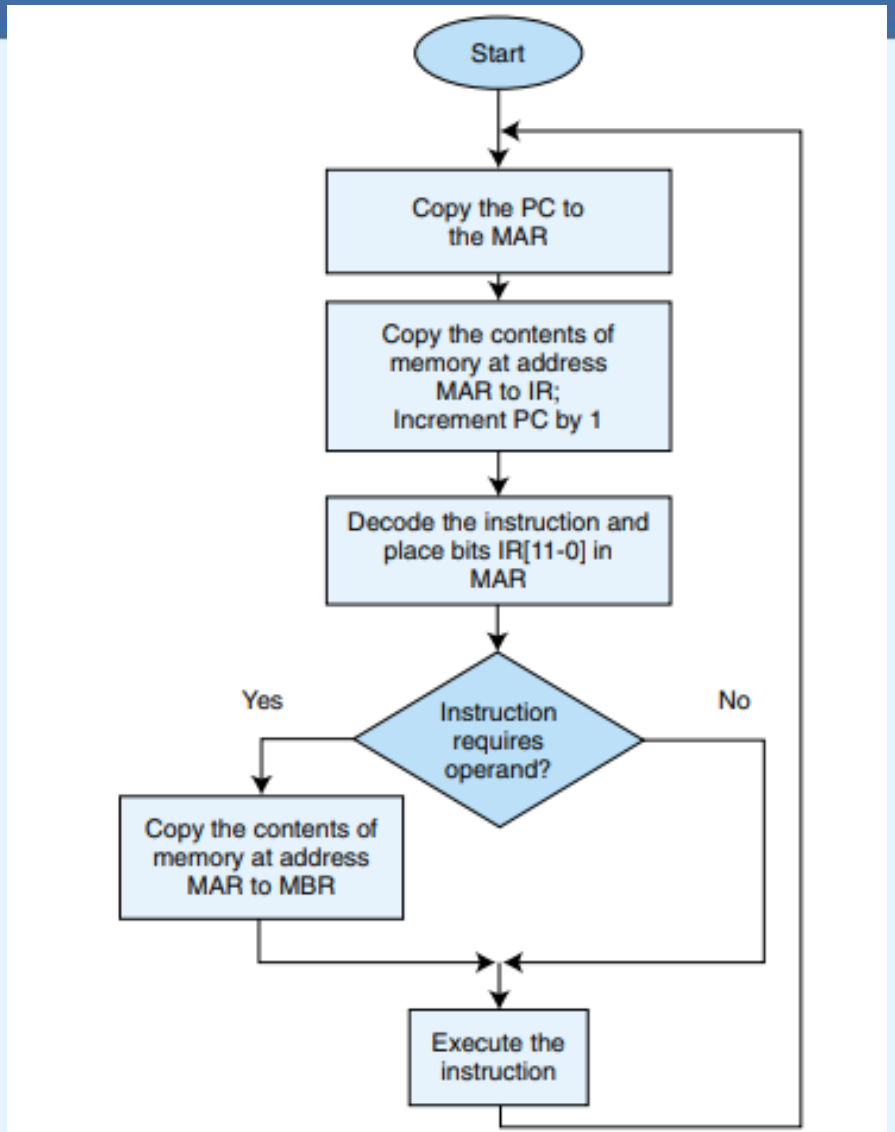
```
If IR[11 - 10] = 00 then
     If AC < 0 then PC ← PC + 1
else If IR[11 - 10] = 01 then
     If AC = 0 then PC ← PC + 1
else If IR[11 - 10] = 10 then
     If AC > 0 then PC ← PC + 1
```

4.3 Instruction Processing

- The *fetch-decode-execute cycle* is the series of steps that a computer carries out when it runs a program.
- We first have to *fetch* an instruction from memory, and place it into the IR.
- Once in the IR, it is *decoded* to determine what needs to be done next.
- If a memory value (operand) is involved in the operation, it is retrieved and placed into the MBR.
- With everything in place, the instruction is *executed*.

The next slide shows a flowchart of this process.

4.3 Instruction Processing

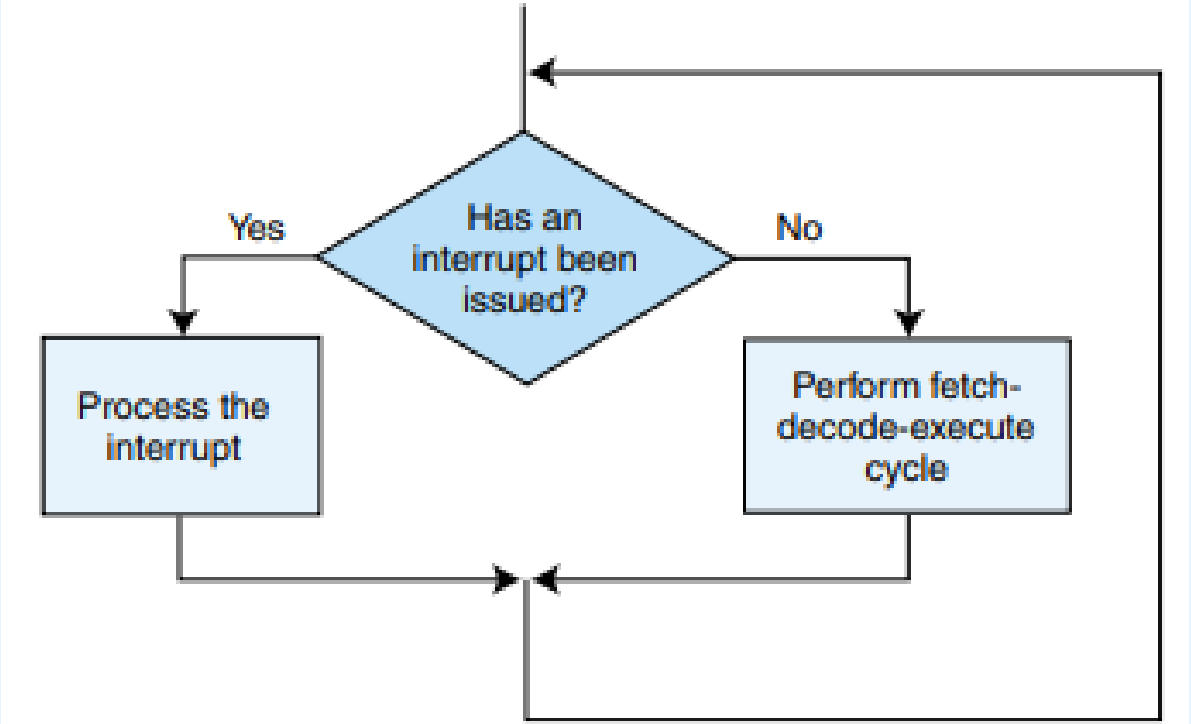


The Fetch-Decode-Execute Cycle

4.3 Instruction Processing

- All computers provide a way of interrupting the fetch-decode-execute cycle.
- Interrupts occur when:
  - A user break (e.,g., Control+C) is issued
  - I/O is requested by the user or a program
  - A critical error occurs
- Interrupts can be caused by hardware or software.
  - Software interrupts are also called *traps*.

4.3 Instruction Processing



Modified Instruction Cycle to Check for Interrupt

4.4 A Simple Program

- Consider the simple MARIE program given below. We show a set of mnemonic instructions stored at addresses 100 - 106 (hex):

Hex Address	Instruction		Binary Contents of Memory Address	Hex Contents of Memory
100	Load	104	0001000100000100	1104
101	Add	105	0011000100000101	3105
102	Store	106	0010000100000110	2106
103	Halt		0111000000000000	7000
104	0023		0000000000100011	0023
105	FFE9		1111111111101001	FFE9
106	0000		0000000000000000	0000

A Program to Add Two Numbers

4.4 A Simple Program

- Let's look at what happens inside the computer when our program runs.
- This is the **`LOAD 104`** instruction:

Step	RTN	PC	IR	MAR	MBR	AC
(initial values)		100	------	------	------	------
Fetch	MAR ← PC	100	------	100	------	------
	IR ← M[MAR]	100	1104	100	------	------
	PC ← PC + 1	101	1104	100	------	------
Decode	MAR ← IR[11–0]	101	1104	104	------	------
	(Decode IR[15–12])	101	1104	104	------	------
Get operand	MBR ← M[MAR]	101	1104	104	0023	------
Execute	AC ← MBR	101	1104	104	0023	0023

4.4 A Simple Program

- Our second instruction is **`ADD 105`**:

Step	RTN	PC	IR	MAR	MBR	AC
(initial values)		101	1104	104	0023	0023
Fetch	MAR ← PC	101	1104	101	0023	0023
	IR ← M[MAR]	101	3105	101	0023	0023
	PC ← PC + 1	102	3105	101	0023	0023
Decode	MAR ← IR[11–0]	102	3105	105	0023	0023
	(Decode IR[15–12])	102	3105	105	0023	0023
Get operand	MBR ← M[MAR]	102	3105	105	FFE9	0023
Execute	AC ← AC + MBR	102	3105	105	FFE9	000C

4.5 A Discussion on Assemblers

- Mnemonic instructions, such as **`LOAD 104`**, are easy for humans to write and understand.
- They are impossible for computers to understand.
- *Assemblers* translate instructions that are comprehensible to humans into the machine language that is comprehensible to computers
  - We note the distinction between an assembler and a compiler: In assembly language, there is a one-to-one correspondence between a mnemonic instruction and its machine code. With compilers, this is not usually the case.

4.5 A Discussion on Assemblers

- Assemblers create an *object program file* from mnemonic *source code* in two passes.
- During the first pass, the assembler assembles as much of the program as it can, while it builds a *symbol table* that contains memory references for all symbols in the program.
- During the second pass, the instructions are completed using the values from the symbol table.

4.5 A Discussion on Assemblers

- Consider our example program (top).
  - Note that we have included two directives **HEX** and **DEC** that specify the radix of the constants.
- During the first pass, we have a symbol table and the partial instructions shown at the bottom.

Address	Instruction
100	Load X
101	Add Y
102	Store Z
103	Halt
X, 104	DEC 35
Y, 105	DEC -23
Z, 106	HEX 0000

X	104
Y	105
Z	106

1 X
3 Y
2 Z
7 0 0 0

4.5 A Discussion on Assemblers

- After the second pass, the assembly is complete.

Address	Instruction
100	Load X
101	Add Y
102	Store Z
103	Halt
X, 104	DEC 35
Y, 105	DEC -23
Z, 106	HEX 0000

1 1 0 4
3 1 0 5
2 1 0 6
7 0 0 0
0 0 2 3
F F E 9
0 0 0 0

X	104
Y	105
Z	106

4.6 Extending Our Instruction Set

- So far, all of the MARIE instructions that we have discussed use a *direct addressing mode*.
- This means that the address of the operand is explicitly stated in the instruction.
- It is often useful to employ an *indirect addressing*, where the address of the address of the operand is given in the instruction.
  - If you have ever used pointers in a program, you are already familiar with indirect addressing.

4.6 Extending Our Instruction Set

Instruction Number (hex)	Instruction	Meaning
0	JnS *X*	Store the PC at address *X* and jump to *X* + 1.
A	Clear	Put all zeros in AC.
B	AddI *X*	Add indirect: Go to address *X*. Use the value at *X* as the actual address of the data operand to add to AC.
C	JumpI *X*	Jump indirect: Go to address *X*. Use the value at *X* as the actual address of the location to jump to.
D	LoadI *X*	Load indirect: Go to address *X*. Use the value at *X* as the actual address of the operand to load into the AC.
E	StoreI *X*	Store indirect: Go to address *X*. Use the value at *X* as the destination address for storing the value in the accumulator.

MARIE's Extended Instruction Set

4.6 Extending Our Instruction Set

- To help you see what happens at the machine level, we have included an indirect addressing mode instruction to the MARIE instruction set.
- The **ADDI** instruction specifies the address of the address of the operand. The following RTL tells us what is happening at the register level:

```
MAR ← X
MBR ← M[MAR]
MAR ← MBR
MBR ← M[MAR]
AC ← AC + MBR
```

4.6 Extending Our Instruction Set

- Similarly

Addl *X*

```
MAR ← X
MBR ← M[MAR]
MAR ← MBR
MBR ← M[MAR]
AC ← AC + MBR
```

Storel X

```
MBR ← X
MBR ← M[MAR]
MAR ← MBR
MBR ← AC
M[MAR] ← MBR
```

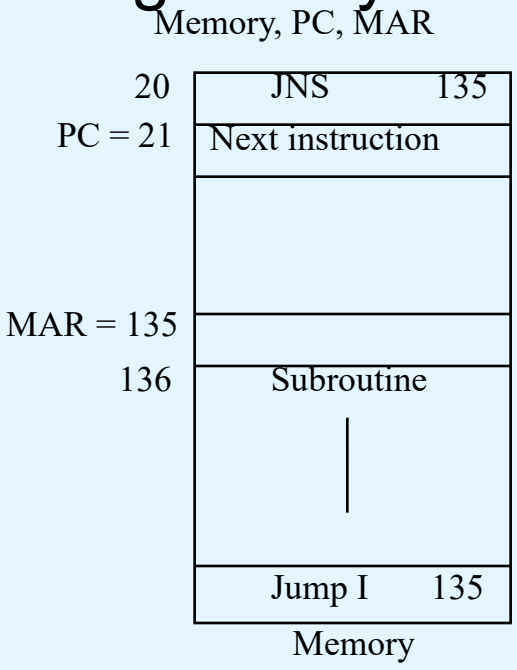
Jumpl *X*

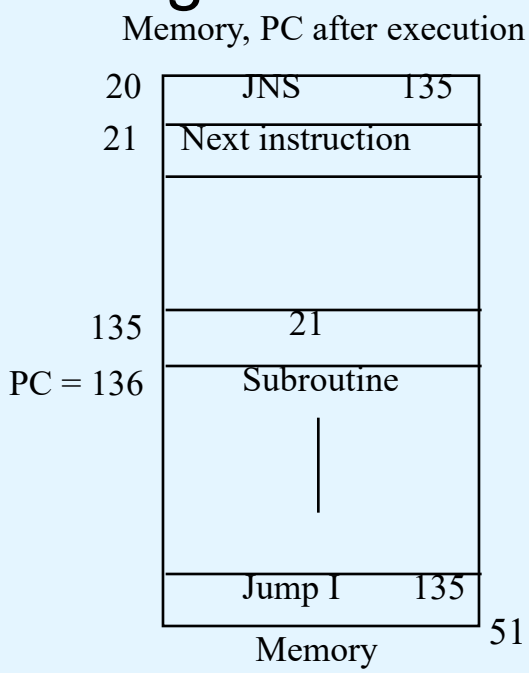
```
MAR ← X
MBR ← M[MAR]
PC ← MBR
```

4.6 Extending Our Instruction Set

- Another helpful programming tool is the use of subroutines.
- The jump-and-store instruction, **JNS**, gives us limited subroutine functionality. The details of the **JNS** instruction are given by the following RTL:

```
MBR ← PC
MAR ← X
M[MAR] ← MBR
MBR ← X
AC ← 1
AC ← AC + MBR
PC ← AC
```





4.6 Extending Our Instruction Set

- Our last helpful instruction is the **CLEAR** instruction.
- All it does is set the contents of the accumulator to all zeroes.
- This is the RTL for **CLEAR**:

$$\mathbf{AC} \leftarrow \mathbf{0}$$

- We put our new instructions to work in the program on the following slide.

Marie's Full Instruction Set

Opcode	Instruction	RTN
0000	JnS *X*	MBR ← PC MAR ← *X* M[MAR] ← MBR MBR ← *X* AC ← 1 AC ← AC + MBR PC ← AC
0001	Load *X*	MAR ← *X* MBR ← M[MAR] AC ← MBR
0010	Store *X*	MAR ← *X*, MBR ← AC M[MAR] ← MBR
0011	Add *X*	MAR ← *X* MBR ← M[MAR] AC ← AC + MBR
0100	Subt *X*	MAR ← *X* MBR ← M[MAR] AC ← AC - MBR
0101	Input	AC ← InREG
0110	Output	OutREG ← AC
0111	Halt	

Marie's Full Instruction Set

Opcode	Instruction	RTN

1000	Skipcond	If IR[11–10] = 00 then
		  If AC < 0 then PC ⟵ PC + 1
		Else If IR[11–10] = 01 then
		  If AC = 0 then PC ⟵ PC + 1
		Else If IR[11–10] = 10 then
		  If AC > 0 then PC ⟵ PC + 1

1001	Jump *X*	PC ⟵ IR[11–0]
1010	Clear	AC ⟵ 0
1011	AddI *X*	MAR ⟵ X MBR ⟵ M[MAR] MAR ⟵ MBR MBR ⟵ M[MAR] AC ⟵ AC + MBR
1100	JumpI *X*	MAR ⟵ *X* MBR ⟵ M[MAR] PC ⟵ MBR
1101	LoadI *X*	MAR ⟵ X MBR ⟵ M[MAR] MAR ⟵ MBR MBR ⟵ M[MAR] AC ⟵ MBR
1110	StoreI *X*	MAR ⟵ X MBR ⟵ M[MAR] MAR ⟵ MBR MBR ⟵ AC M[MAR] ⟵ MBR

EXAMPLE 4.1

Here is an example using a loop to add five numbers:

```
100 |       LOAD Addr
101 |       STORE Next
102 |       LOAD Num
103 |       SUBT One
104 |       STORE Ctr
105 |       CLEAR
106 |Loop   LOAD Sum
107 |       ADDI Next
108 |       STORE Sum
109 |       LOAD Next
10A |       ADD One
10B |       STORE Next
10C |       LOAD Ctr
10D |       SUBT One
```

```
10E |       STORE Ctr
10F |       SKIPCOND 000
110 |       JUMP Loop
111 |       HALT
112 |Addr   HEX 118
113 |Next   HEX 0
114 |Num    DEC 5
115 |Sum    DEC 0
116 |Ctr    HEX 0
117 |One    DEC 1
118 |       DEC 10
119 |       DEC 15
11A |       DEC 2
11B |       DEC 25
11C |       DEC 30
```

```
Address         Instruction
100             Load      Addr    /Load address of first number to be added
101             Store     Next    /Store this address as our Next pointer
102             Load      Num     /Load the number of items to be added
103             Subt      One     /Decrement
104             Store     Ctr     /Store this value in Ctr to control looping
105     Loop,   Load      Sum     /Load the Sum into AC
106             AddI      Next    /Add the value pointed to by location Next
107             Store     Sum     /Store this sum
108             Load      Next    /Load Next
109             Add       One     /Increment by one to point to next address
10A             Store     Next    /Store in our pointer Next
10B             Load      Ctr     /Load the loop control variable
10C             Subt      One     /Subtract one from the loop control variable
10D             Store     Ctr     /Store this new value in loop control variable
10E             Skipcond  000     /If control variable < 0, skip next
                                  /instruction
10F             Jump      Loop    /Otherwise, go to Loop
110             Halt              /Terminate program
111     Addr,   Hex       117     /Numbers to be summed start at location 117
112     Next,   Hex       0       /A pointer to the next number to add
113     Num,    Dec       5       /The number of values to add
114     Sum,    Dec       0       /The sum
115     Ctr,    Hex       0       /The loop control variable
116     One,    Dec       1       /Used to increment and decrement by 1
117             Dec       10      /The values to be added together
118             Dec       15
119             Dec       20
11A             Dec       25
11B             Dec       30
```

Note: Line numbers in program are given for information only and are not used in the MarieSim environment.

EXAMPLE 4.2:

- This example illustrates the use of an if/else construct to allow for selection. It implements the following:

```
if X = Y then
        X = X *2
else
        Y = Y - X;
```

EXAMPLE 4.2:

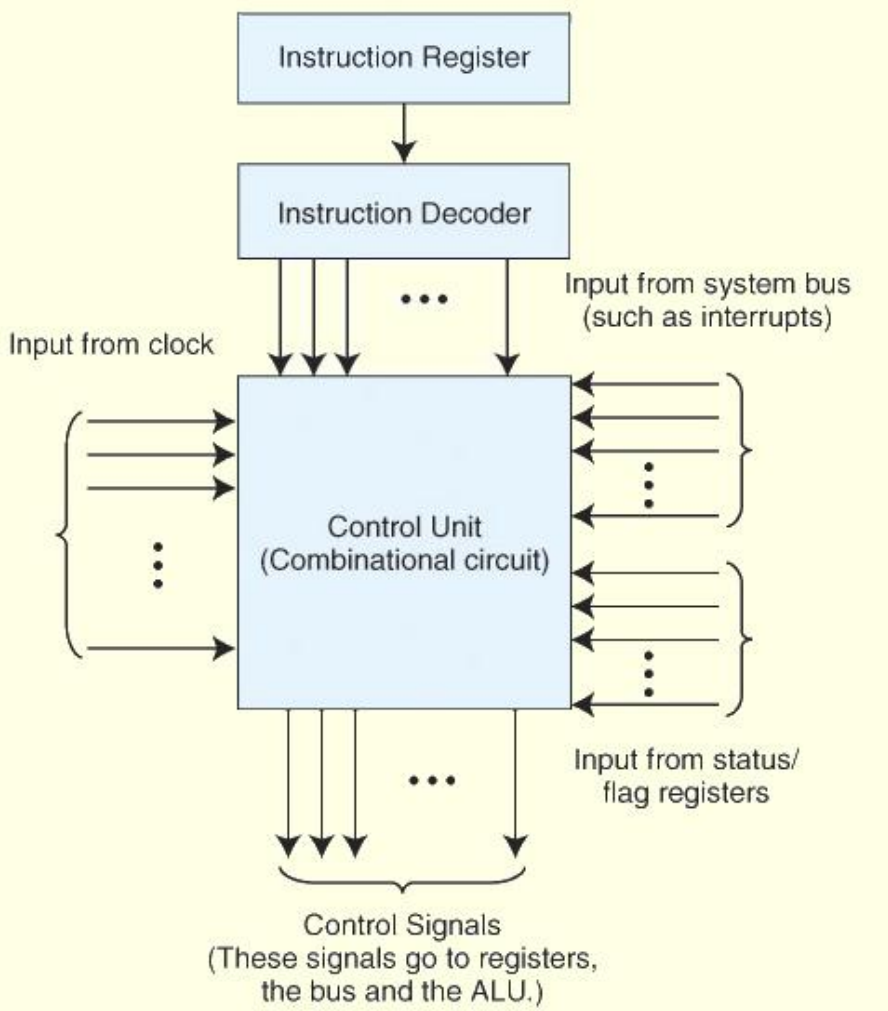
```
Address          Instruction
100       If,    Load       X       /Load the first value
101              Subt       Y       /Subtract the value of Y, store result in AC
102              Skipcond   400     /If AC = 0, skip the next instruction
103              Jump       Else    /Jump to Else part if AC is not equal to 0
104       Then,  Load       X       /Reload X so it can be doubled
105              Add        X       /Double X
106              Store      X       /Store the new value
107              Jump       Endif   /Skip over the false, or else, part to end of
                                    /if
108       Else,  Load       Y       /Start the else part by loading Y
109              Subt       X       /Subtract X from Y
10A              Store      Y       /Store Y - X in Y
10B       Endif, Halt               /Terminate program (it doesn't do much!)
10C       X,     Dec        12      /Load the loop control variable
10D       Y,     Dec        20      /Subtract one from the loop control variable
```

4.7 A Discussion on Decoding

- A computer's control unit keeps things synchronized, making sure that bits flow to the correct components as the components are needed.
- There are two general ways in which a control unit can be implemented: *hardwired control* and *microprogrammed control*.
  - With microprogrammed control, a small program is placed into read-only memory in the microcontroller.
  - Hardwired controllers implement this program using digital logic components.

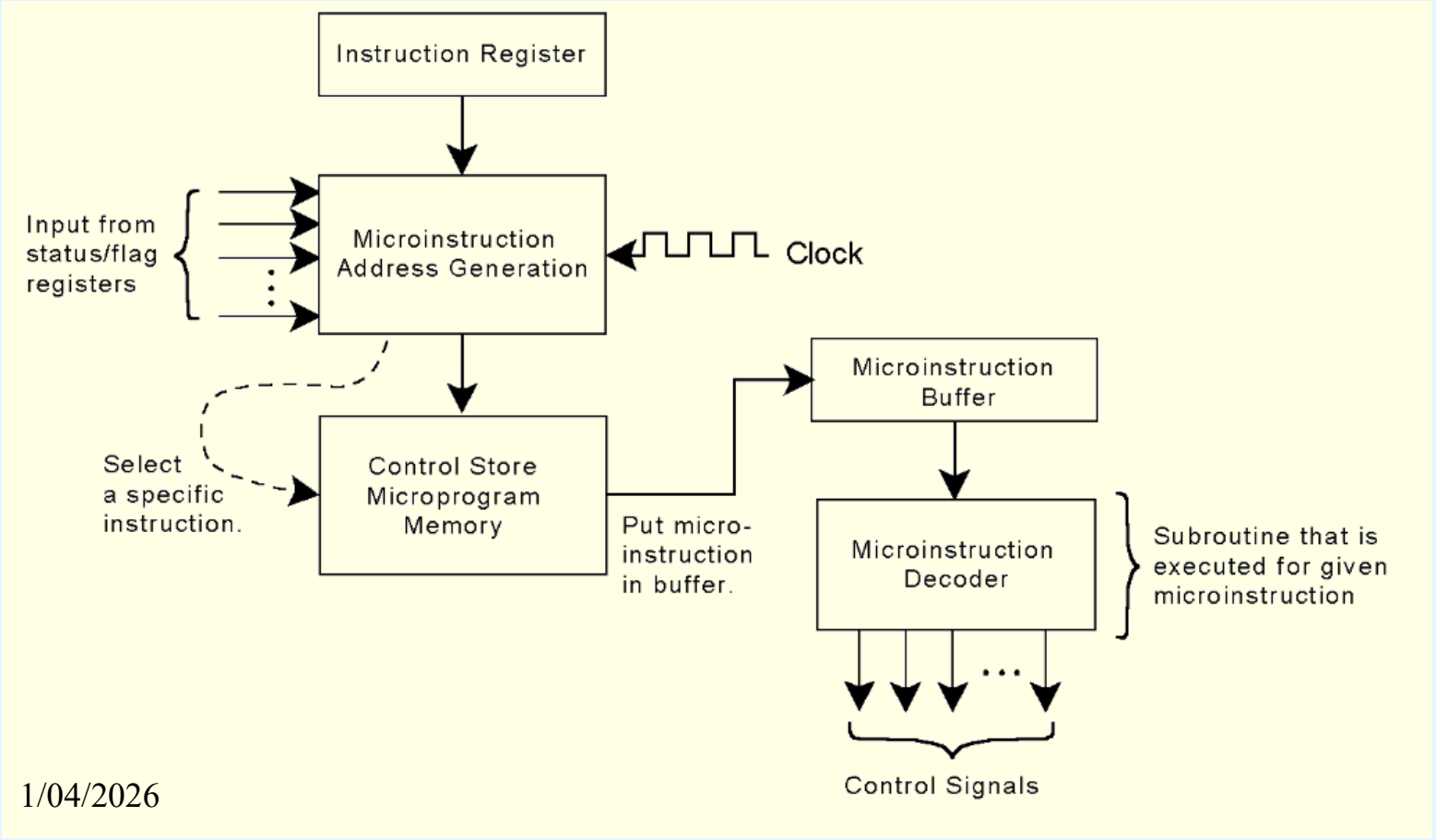
4.7 A Discussion on Decoding

- For example, a hardwired control unit for our simple system would need a 4-to-14 decoder to decode the opcode of an instruction.
- The block diagram at the right, shows a general configuration for a hardwired control unit.



4.7 A Discussion on Decoding

- In microprogrammed control, the *control store* is kept in ROM, PROM, or EPROM firmware, as shown below.



4.8 Real World Architectures

- MARIE shares many features with modern architectures, but it is not an accurate depiction of them.
- In the following slides, we briefly examine two machine architectures.
- We will look at an Intel architecture, which is a CISC machine and MIPS, which is a RISC machine.
  - CISC is an acronym for complex instruction set computer.
  - RISC stands for reduced instruction set computer.
  - MIPS (Microprocessor without Interlocked Pipelined Stages)

4.8 Real World Architectures

- The classic Intel architecture, the 8086, was born in 1979. It is a CISC architecture.
- It was adopted by IBM for its famed PC, which was released in 1981.
- The 8086 operated on 16-bit data words and supported 20-bit memory addresses.
- Later, to lower costs, the 8-bit 8088 was introduced. Like the 8086, it used 20-bit memory addresses.

What was the largest memory that the 8086 could address?

4.8 Real World Architectures

- The 8086 had four 16-bit general-purpose registers that could be accessed by the half-word.
- It also had a flags register, an instruction register, and a stack accessed through the values in two other registers, the base pointer and the stack pointer.
- The 8086 had no built in floating-point processing.
- In 1980, Intel released the 8087 numeric coprocessor, but few users elected to install them because of their cost.

4.8 Real World Architectures

- In 1985, Intel introduced the 32-bit 80386.
- It also had no built-in floating-point unit.
- The 80486, introduced in 1989, was an 80386 that had built-in floating-point processing and cache memory.
- The 80386 and 80486 offered downward compatibility with the 8086 and 8088.
- Software written for the smaller word systems was directed to use the lower 16 bits of the 32-bit registers.

4.8 Real World Architectures

- Currently, Intel's most advanced 32-bit microprocessor is the Pentium 4.
- It can run as fast as 3.06 GHz. This clock rate is over 350 times faster than that of the 8086.
- Speed enhancing features include multilevel cache and instruction pipelining.
- Intel, along with many others, is marrying many of the ideas of RISC architectures with microprocessors that are largely CISC.

4.8 Real World Architectures

- The MIPS family of CPUs has been one of the most successful in its class.
- In 1986 the first MIPS CPU was announced.
- It had a 32-bit word size and could address 4GB of memory.
- Over the years, MIPS processors have been used in general purpose computers as well as in games.
- The MIPS architecture now offers 32- and 64-bit versions.

4.8 Real World Architectures

- MIPS was one of the first RISC microprocessors.
- The original MIPS architecture had only 55 different instructions, as compared with the 8086 which had over 100.
- MIPS was designed with performance in mind: It is a *load/store* architecture, meaning that only the load and store instructions can access memory.
- The large number of registers in the MIPS architecture keeps bus traffic to a minimum.

How does this design affect performance?

Chapter 4 Conclusion

- The major components of a computer system are its control unit, registers, memory, ALU, and data path.
- A built-in clock keeps everything synchronized.
- Control units can be microprogrammed or hardwired.
- Hardwired control units give better performance, while microprogrammed units are more adaptable to changes.

Chapter 4 Conclusion

- Computers run programs through iterative fetch-decode-execute cycles.
- Computers can run programs that are in machine language.
- An assembler converts mnemonic code to machine language.
- The Intel architecture is an example of a CISC architecture; MIPS is an example of a RISC architecture.